

Zaibin Zhang

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Google Scholar
Ph.D. Candidate · Embodied AI / VLA / MLLM

Education
University of Oxford
Sept. 2025 – Mar. 2026
Visiting Ph.D. Student · Supervisor: Prof. Philip Torr
Dalian University of Technology
Sept. 2022 – Present
Ph.D. in Signal & Information Processing · Advisor:
Prof. Huchuan Lu (IEEE Fellow)
Dalian University of Technology
Sept. 2019 – Jun. 2022
M.S. in Mechanical Engineering · National Scholarship



Research Direction

My research centers on **Agentic Embodied AI**, aiming to integrate the reasoning, memory, and interaction capabilities of multimodal agents with real-world robotic execution. Rather than relying solely on demonstration scaling, I study **compositional intelligence**: how robots can decompose complex tasks, reuse learned skills, coordinate across different arms, accumulate experience over long-horizon interactions, and recombine existing capabilities to solve open-ended tasks without retraining. This direction builds on my work in multimodal and embodied agents, including Think3D, SPAgent, MA-VLA, and MACG-Bench. In parallel, OASIS—an open platform for million-scale agent societies with nearly 5k GitHub stars—reflects my strong open-source practice, while my early work on multi-agent safety, recognized with an **ACL Outstanding Paper Award**, demonstrates my ability to identify important research problems at an early stage.

Research Experience

University of Oxford · Visiting Ph.D. Student · Supervisor: Prof. Philip Torr
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MA-VLA: Multi-Arm Vision-Language-Action Model for Collaboration and Compositional Generalization · VLA;Multi-Arm;Arm Shuf- 2025
fle;Compositional Generalization;Embodied AI
ECCV 2026 · First Author

- Background:** Most multi-arm VLAs read language as one static command and do not make clear per-arm assignments, so they often work on familiar routines but break on new coordination patterns. We define *multi-arm compositional generalization*: reusing known atomic actions to solve unseen collaborative tasks without retraining.
- Method:** MA-VLA turns a high-level instruction into interpretable *atomic action sequences* and assigns them to each arm in one model. *Arm Shuffle* randomly changes the arm-observation/state/prompt correspondence during training, so each arm learns reusable, role-agnostic skills instead of memorizing a fixed role.
- Results:** Outperforms Pi0, Pi0-FAST, and DP3 on RoboFactory (2–4 arms, **83.3% avg. 4-arm**), RoboTwin 2.0 (Hard), and real-world dual-arm SO101. It is the first model in our setting to obtain non-zero OOD compositional generalization where prior VLAs score 0%.

Attention-DP3: Spatially Object-aware 3D Diffusion Policy via Geometry-aligned Attentional Conditioning · 3D Diffusion Policy;Point 2025
Cloud;Tri-Field Attention;Clutter Robustness;Embodied AI
ECCV 2026 · Co-First Author

- Background:** In cluttered scenes, a point cloud alone often makes it hard to tell which 3D points belong to the language-specified object, especially with occlusion or nearby distractors. Standard 3D diffusion policies degrade quickly as clutter increases.
- Method:** Proposed *Tri-Field Attentional Conditioning*: lift open-vocabulary 2D object masks into 3D with calibrated camera geometry, then build three fields for target anchoring, within-target saliency, and background suppression. This adds object-level guidance without changing the DP3 diffusion backbone.
- Results:** Improves over DP3 on Adroit, DexArt, MetaWorld, and real-world SO101; reaches **up to +31% over DP3 under heavy clutter (5+ distractors)** and remains stable when the DP3 baseline drops.

MACG-Bench: Benchmarking Compositional Generalization in Multi-Arm Manipulation · Benchmark;Multi-Arm;Compositional 2025
Generalization;VLA;AE-VLA
Under Review · First Author

- Background:** Many multi-arm evaluations test fixed collaboration routines, so good scores can still come from memorizing familiar coordination templates. MACG-Bench is designed to test whether a model can actually recombine skills in new multi-arm settings.
- Contributions:** Built **MACG-Bench** with 8 training tasks over 10 atomic skills and 23 test tasks covering new execution orders, synchronization patterns, and subtask combinations. Also proposed **AE-VLA**, using arm-specific tokens and a global collaboration token for structured cross-arm communication, with **40%+ improvement** in compositional generalization success rate.
- Key Finding:** Current VLA policies handle seen routines much better than unseen collaboration structures, suggesting that compositional skill reuse is still a major bottleneck for embodied models.

Think3D: Thinking with Space for Spatial Reasoning · VLM;3D Spatial Reasoning;Active Exploration;RL;Training-Free Jan. 2026
arXiv 2601.13029 · First Author · Arxiv · Code · 190+ Stars

- Background:** Most VLMs reason from fixed 2D images, while many spatial questions are easier if the agent can move its viewpoint and inspect the scene in 3D. We study how to give VLM agents this kind of active spatial reasoning ability.
- Framework (Think3D):** Built a framework for *interactive 3D chain-of-thought* reasoning, combining 3D reconstruction (point clouds, camera poses), ego/global-view switching, and camera-based spatial manipulation so the model can actively explore the scene.
- Training-Free Gains:** Zero-shot plug-in for closed-source VLMs (GPT-4.1, Gemini 2.5 Pro): **+7.8%** on BLINK MV/ MindCube, **+4.7%** on VSI-Bench.
- Key Discovery (Think3D-RL):** RL-trained smaller open-weight models learn similar 3D exploration patterns to large proprietary models (GPT-4.1, Gemini 2.5 Pro), raising Qwen3-VL-4B gains from **+0.7% to +10.7%** without trajectory annotations.

Open-Source Projects & Award-Winning Papers

PsySafe (ACL 2024 · First Author · **Outstanding Paper Award** · Cited 116)—first systematic analysis of multi-agent system safety, covering attack, defense, and evaluation; later papers studying the safety of agent systems such as Claude Code and OpenClaw draw on this problem framing.

OASIS (NeurIPS-W 2024 · Technical Report · Co-First Author · **4800+ Stars** · Cited 185)—bet early on agent societies and built the first open-source base framework to simulate them at million-agent scale; now foundational infrastructure for the field and the basis of the Mirofish venture (**60k+ Stars**, ¥30M seed round).

CoMAS (ICLR 2026 · Core Member)—studied how agents can improve through interaction by using peer feedback as intrinsic rewards, forming a self-supervised RL loop for multi-agent co-evolution.

Honors & Awards

Outstanding Paper Award · The 62nd Annual Meeting of the Association for Computational Linguistics (ACL 2024) · Bangkok, Thailand 2024

National Scholarship · Dalian University of Technology · Dalian, China 2021

Technical Skills

- Embodied AI & Robotics** VLA training and evaluation; diffusion policies (DP, DP3) and flow-matching control; multi-arm collaboration and compositional generalization; configured UMI with new cameras / GoPro 13 and built the dual-arm UR5e data-collection-to-VLA pipeline.
- Multimodal LLMs** MLLM/VLM fine-tuning and RL training (SFT, GRPO); 3D spatial reasoning, point-cloud encoding, and 2D–3D geometric lifting; agent reasoning and tool-use workflows.
- Training & Engineering** HuggingFace, Swift, LeRobot, vLLM, verl, OpenRLHF; Python / PyTorch / CUDA.